

DS & BDA Lab - Viva Questions & Answers

GROUP A: Assignment 2 – MapReduce (Viva Questions)

1. What is use of Reporter?

Reporter is an interface in Hadoop MapReduce used to report the progress of a task and update status messages to the JobTracker. It also allows the mapper or reducer to send counters and status strings back to the framework during long-running tasks.

2. How to put input file in HDFS?

Use the command: `hadoop fs -put <local_file_path> <hdfs_destination_path>`. For example, `hadoop fs -put input.txt /user/hadoop/input/` copies a local file into HDFS so it can be processed by MapReduce jobs.

3. Explain what is Speculative Execution?

Speculative Execution is a Hadoop feature where if a task is running slower than expected, the framework launches a duplicate (speculative) copy of the same task on another node. Whichever copy finishes first is used, and the other is killed, helping reduce job completion time.

4. Explain what are the basic parameters of a Mapper?

A Mapper in Hadoop takes four parameters: input key (LongWritable), input value (Text), output key, and output value. The `map()` method processes each input record and emits intermediate key-value pairs that are then passed to the Reducer.

5. Explain what is the function of MapReducer practitioner?

A MapReduce practitioner designs, implements, and optimizes distributed data processing applications using the MapReduce programming model. They write Mapper and Reducer classes to process large datasets efficiently across a Hadoop cluster.

6. Explain what is difference between an Input Split and HDFS Block?

An HDFS Block is the physical chunk of data stored on disk (default 128 MB), while an Input Split is the logical division of data that a single Mapper processes. An Input Split can span multiple HDFS blocks, but by default they are aligned to avoid network overhead.

7. Explain how JobTracker schedules a task?

JobTracker receives a job from the client, splits it into map and reduce tasks, and assigns them to TaskTrackers based on data locality and resource availability. It monitors task progress and reschedules failed tasks on other nodes to ensure job completion.

GROUP A: Assignment 3 – HiveQL & HBase (Viva Questions)

8. Explain about SORT BY, ORDER BY, DISTRIBUTE BY and CLUSTER BY in Hive.

ORDER BY performs a global sort of all data using a single reducer. SORT BY sorts within each reducer locally. DISTRIBUTE BY controls how data is distributed among reducers based on a column.

CLUSTER BY is a combination of DISTRIBUTE BY and SORT BY, distributing and sorting data using the same columns.

9. What is the difference between HBase and Hive?

HBase is a NoSQL, real-time read/write database built on HDFS designed for random access of large datasets. Hive is a data warehouse tool that provides SQL-like querying (HiveQL) over HDFS for batch processing and analytical queries, not real-time operations.

10. What is the default database provided by Hive for Metastore?

By default, Hive uses Apache Derby as its embedded metastore database. For production use, it is recommended to configure an external relational database like MySQL or PostgreSQL as the metastore to support concurrent access.

11. What is Apache HBase and its use?

Apache HBase is a distributed, scalable, NoSQL database that runs on top of HDFS. It is modeled after Google's Bigtable and is used for storing and retrieving large amounts of sparse data with low-latency random read/write access in real time.

12. Give the name of the key components of HBase.

The key components of HBase are: HMaster (master server that manages region assignments), RegionServer (handles read/write requests for regions), ZooKeeper (coordination service), HLog (write-ahead log), and MemStore/HFile (in-memory and disk-based storage).

13. What is the use of get() method?

The get() method in HBase is used to retrieve a specific row from a table using its row key. It returns the cells (column family, column qualifier, value, timestamp) for the specified row, allowing precise, single-record lookups from the HBase table.

14. What is the reason of using HBase?

HBase is used when applications require real-time, random read/write access to very large datasets (billions of rows, millions of columns). It provides horizontal scalability, automatic sharding, fault tolerance via HDFS, and suits use cases like messaging systems, time-series data, and user profile storage.

15. Define column families in HBase?

Column families in HBase are groups of related columns stored together on disk for performance optimization. All columns within a family share the same storage properties and are defined at table creation time. A column family name is followed by a column qualifier to identify a specific column.

16. Define standalone mode in HBase?

Standalone mode is the default HBase configuration where all HBase daemons (HMaster, RegionServer, ZooKeeper) run within a single JVM on the local machine. It uses the local file system instead of HDFS and is suitable for development and testing purposes.

17. What is RegionServer?

A RegionServer is a worker node in HBase that hosts one or more Regions (subsets of a table's rows). It handles all read and write requests for its assigned regions, manages MemStores, performs compactions, and communicates with the HMaster for region assignments.

18. What is the use of MasterServer and HMaster?

HMaster is the master daemon in HBase responsible for coordinating RegionServers, assigning regions to RegionServers, handling DDL operations (create/delete tables), and monitoring cluster health. It does not serve client data requests directly but manages the overall cluster metadata.

19. What are the operational commands of HBase?

Key HBase shell commands include: create (create table), list (list tables), put (insert data), get (retrieve row), scan (scan table), delete (delete data), disable/enable (manage table state), describe (show table schema), and drop (delete table).

20. Which command is used to run HBase Shell?

The command `hbase shell` is used to start the HBase interactive shell. This launches a JRuby-based REPL where users can execute HBase commands to manage tables, insert data, query records, and administer the HBase cluster.

21. What is the use of ZooKeeper?

ZooKeeper is a distributed coordination service used by HBase to manage cluster state. It tracks which RegionServers are alive, stores the location of the ROOT and META tables, coordinates distributed locks, and ensures consistency and synchronization across the HBase cluster.

GROUP B: Assignment 4 – Python Data Operations (Viva Questions)

22. What is the purpose of creating data subsets in a dataset?

Data subsets allow analysts to focus on specific portions of a dataset relevant to a particular analysis, reducing computational overhead and improving clarity. They help in isolating variables of interest, filtering noise, and making datasets more manageable for exploration and modeling.

23. How would you select specific rows and columns in Python using pandas?

Use `df.loc[]` for label-based selection (e.g., `df.loc[0:5, ['col1', 'col2']]`) and `df.iloc[]` for integer-position-based selection. For conditional row filtering, use boolean indexing like `df[df['column'] > value]` to select rows meeting a specific condition.

24. How can subsets be helpful in exploratory data analysis?

Subsets help in understanding distributions and relationships within specific groups or categories of data. They allow analysts to compare patterns between segments (e.g., high vs. low engagement posts), making it easier to derive targeted insights without processing the entire dataset.

25. What are the different types of merges (joins) in pandas, and how do they differ?

Pandas supports four join types: inner join (only matching rows from both DataFrames), outer join (all rows from both, NaN for non-matches), left join (all rows from left DataFrame), and right join (all rows from right DataFrame). They differ in how unmatched records are handled.

26. How do you handle conflicts when merging datasets with overlapping column names?

Pandas automatically appends suffixes (`_x` and `_y`) to overlapping column names during a merge. You can customize this using the `suffixes` parameter in `pd.merge()`, e.g., `suffixes=('_left', '_right')`, to make the resulting column names more meaningful and avoid confusion.

27. What is the difference between sorting data by rows and by columns?

Sorting by rows (`sort_values()`) rearranges the order of records based on values in one or more columns, useful for ranking or ordering data. Sorting by columns (`sort_index(axis=1)`) reorders the column headers alphabetically or by a specified key, useful for organizing feature presentation.

28. Can you explain how sorting can be helpful in analyzing metrics like post engagement?

Sorting by engagement metrics (likes, shares, comments) in descending order quickly identifies top-performing posts. This helps marketers and analysts understand what content resonates best with audiences, enabling data-driven decisions for future content strategy.

29. What does transposing data mean, and when is it useful?

Transposing converts rows to columns and columns to rows using `df.T` or `df.transpose()`. It is useful when features are stored as rows instead of columns, when comparing records side-by-side, or when preparing data for libraries that expect a specific orientation.

30. Are there any limitations to transposing large datasets?

Transposing very large datasets can consume significant memory since a full copy of the data is created with swapped dimensions. Also, data type consistency may be lost if columns have mixed types, as all values in a transposed row must share a single dtype, often converting to object.

31. What is the difference between the shape and size of a dataset?

`df.shape` returns a tuple (rows, columns) indicating the dataset's dimensions. `df.size` returns the total number of elements (rows x columns). Shape gives structural information while size gives the total count of cells, including those containing NaN or null values.

32. How would you reshape a dataset from wide to long format or vice versa in Python?

Use `pd.melt()` to convert wide format (multiple columns) to long format (key-value pairs). Use `df.pivot()` or `df.pivot_table()` to convert long format back to wide format. These operations are essential when preparing data for specific visualization libraries or machine learning models.

33. What is the importance of reshaping data when preparing it for analysis or visualization?

Many visualization tools (like seaborn) and machine learning algorithms expect data in a specific format (long or wide). Reshaping ensures compatibility, reduces redundancy, and makes it easier to apply group operations, aggregations, and statistical transformations on the dataset.

34. Why is it essential to preprocess the Facebook Metrics dataset before analysis?

The raw Facebook Metrics dataset may contain missing values, inconsistent data types, and irrelevant columns that can skew analysis results. Preprocessing ensures data quality, standardizes formats, and prepares the dataset for accurate statistical analysis and machine learning model training.

GROUP B: Assignment 5 – Data Cleaning, Transformation & Model Building (Viva Questions)

35. What are the common methods to handle missing data in a dataset?

Missing data can be handled by imputation (replacing with mean, median, or mode), dropping rows/columns with excessive missing values using `dropna()`, or using forward/backward fill for time-series data. The choice depends on the percentage of missing values and their impact on the analysis.

36. Why is it important to remove duplicate rows from a dataset?

Duplicate rows can bias statistical analysis by over-representing certain records, leading to skewed distributions, inflated model accuracy, and incorrect insights. Removing duplicates using `df.drop_duplicates()` ensures each observation is counted once, maintaining data integrity and model fairness.

37. What challenges can arise during data integration from multiple sources?

Challenges include mismatched column names or data types, different scales or units across datasets, conflicting values for the same entity, and missing common keys for joining. These issues require careful schema mapping, type casting, and validation to produce a consistent, reliable integrated dataset.

38. How do you ensure data consistency when merging datasets?

Ensure common key columns have the same data type and format before merging. After merging, validate the row count against expectations, check for unexpected NaN values, and verify that merged values are logically consistent. Using outer joins and then inspecting unmatched records helps identify inconsistencies.

39. Why do we standardize or normalize data before feeding it into a model?

Standardization (zero mean, unit variance) and normalization (scaling to 0-1 range) ensure that no single feature dominates the model due to its larger scale. This is critical for distance-based algorithms (KNN, SVM) and gradient descent-based models where feature scale directly affects convergence and performance.

40. What is the difference between one-hot encoding and label encoding?

Label encoding assigns a unique integer to each category (e.g., Red=0, Blue=1), which may imply a false ordinal relationship. One-hot encoding creates binary columns for each category (e.g., is_Red, is_Blue), avoiding ordinal assumptions and is preferred for nominal categorical variables in most ML models.

41. How can you identify anomalies in a dataset?

Anomalies can be detected using statistical methods like Z-score (values beyond 3 standard deviations) or IQR (values below $Q1 - 1.5 * IQR$ or above $Q3 + 1.5 * IQR$). Visual methods like box plots, scatter plots, and histograms also help identify outliers and unusual patterns in the data.

42. What techniques are used to correct outliers in numerical data?

Outliers can be removed (dropping rows), capped (Winsorization - replacing with 5th/95th percentile values), or transformed (log/square root transformation to reduce skewness). The appropriate technique depends on whether outliers are genuine data errors or meaningful extreme observations.

43. How do you evaluate the performance of a classification model?

Classification models are evaluated using metrics like accuracy (correct predictions/total), precision (true positives / predicted positives), recall (true positives / actual positives), F1-score (harmonic mean of precision and recall), and ROC-AUC score which measures the model's discriminative ability.

44. Can you explain the difference between training and testing datasets?

The training dataset is used to fit the model, allowing it to learn patterns and relationships from the data. The testing dataset is a held-out subset used to evaluate model performance on unseen data. A typical split is 80:20 (train:test) to assess generalization ability and detect overfitting.

45. How does air quality impact heart diseases, and why is it essential to study this relationship?

Poor air quality (high particulate matter, PM2.5, NO2 levels) contributes to cardiovascular inflammation, atherosclerosis, and increased risk of heart attacks. Studying this relationship using integrated datasets helps establish causal links, informing public health policies and preventive healthcare measures.

46. What are the key features that you would prioritize in predicting heart disease risk?

Key features include age, cholesterol level, blood pressure (resting and peak), chest pain type, maximum heart rate, exercise-induced angina, and ST depression. These are clinically significant predictors identified through medical research and feature importance analysis in machine learning models.

GROUP B: Assignment 6 – Data Visualization with Matplotlib & Seaborn (Viva Questions)

47. What is the importance of data visualization in data analysis?

Data visualization transforms complex numerical data into intuitive graphical representations, making it easier to identify patterns, trends, correlations, and outliers that may not be apparent in raw data. It facilitates faster decision-making, effective communication of findings, and supports exploratory data analysis.

48. How do Matplotlib and Seaborn differ in terms of functionality and ease of use?

Matplotlib provides low-level, highly customizable plotting control requiring more code for styling. Seaborn is built on Matplotlib and provides a higher-level interface with better default aesthetics and built-in support for statistical visualizations (heatmaps, violin plots, pairplots) with minimal code.

49. What types of graphs are most suitable for showing trends over time?

Line charts are the most suitable for showing trends over time as they connect data points sequentially, making temporal patterns and directional changes clearly visible. Area charts can also be used when showing cumulative values, while bar charts work for discrete time intervals like months or years.

50. Why is it important to choose the right type of plot for different types of data?

Using an inappropriate plot can mislead viewers and hide important patterns. For example, a pie chart for time-series data is ineffective, while a bar chart cannot show distribution shape. The right plot type ensures the data's story is communicated accurately and the visualization is easy to interpret.

51. How do you customize the appearance (colors, labels, gridlines) of a plot in Matplotlib?

Use `plt.color()` or the `color` parameter in plot functions for colors. `plt.xlabel()`, `plt.ylabel()` set axis labels. `plt.grid(True)` adds gridlines. `plt.title()` sets the title. The `plt.rcParams` dictionary allows global style customization, and `plt.style.use()` applies predefined themes like 'ggplot' or 'seaborn'.

52. How would you add a title and labels to the axes of a plot in Matplotlib?

Use `plt.title('Title Text')` to add a plot title, `plt.xlabel('X-axis Label')` for the x-axis, and `plt.ylabel('Y-axis Label')` for the y-axis. For subplots, use `ax.set_title()`, `ax.set_xlabel()`, and `ax.set_ylabel()` on the specific Axes object for precise control.

53. How can you save a plot as an image or PDF in Matplotlib?

Use `plt.savefig('filename.png', dpi=300, bbox_inches='tight')` to save as PNG with high resolution. Replace `'png'` with `'pdf'` for PDF format or `'svg'` for vector format. The `bbox_inches='tight'` parameter ensures all labels and titles are included in the saved file without clipping.

54. What advantages does Seaborn offer over Matplotlib for statistical visualizations?

Seaborn automatically computes and displays statistical estimates (confidence intervals, regression lines), integrates seamlessly with Pandas DataFrames, provides built-in themes with attractive aesthetics, and simplifies complex plots like heatmaps, pairplots, and violin plots that require extensive code in Matplotlib.

55. How can you plot a heatmap using Seaborn, and what kind of data is suitable for a heatmap?

Use `sns.heatmap(df.corr(), annot=True, cmap='coolwarm')` to plot a correlation heatmap. Heatmaps are suitable for matrix-type data showing relationships between variables, such as correlation matrices, confusion matrices, or any 2D grid where color intensity represents the magnitude of values.

56. Can you explain how Seaborn's FacetGrid works and when to use it?

FacetGrid creates a grid of subplots based on the values of one or more categorical variables, allowing comparison of plot patterns across groups. Use it when you want to visualize how a relationship between variables changes across different categories, like comparing heart disease rates by age and gender simultaneously.

57. How do you visualize categorical data effectively in Seaborn?

Use `sns.countplot()` for frequency counts, `sns.barplot()` for mean values with confidence intervals, `sns.boxplot()` for distribution across categories, and `sns.violinplot()` for distribution shape. `sns.catplot()` provides a unified interface for all categorical plots with the `type` parameter.

58. How would you handle missing or outlier data when creating visualizations?

Drop or impute missing values before plotting to avoid NaN-related errors. For outliers, either remove them if they are errors or use robust plots like box plots that explicitly show outliers. Adding data range limits with `plt.xlim()` or `plt.ylim()` can also prevent outliers from distorting the visual scale.

GROUP B: Assignment 7 – Tableau Data Visualization (Viva Questions)

59. Why is data visualization important in the analysis of datasets like Adult and Iris?

Visualization helps uncover hidden patterns, class distributions, feature correlations, and outliers in complex datasets. For the Iris dataset, scatter plots reveal clear species clusters, while for the Adult dataset, bar and heatmap charts expose income distribution across education and occupation categories.

60. How do you choose the most appropriate visualization type for different data attributes?

For continuous variables, use histograms, scatter plots, or box plots. For categorical variables, use bar charts or count plots. For relationships between two continuous variables, use scatter plots or heatmaps. For hierarchical data, use treemaps. The data type (nominal, ordinal, interval, ratio) guides the choice.

61. What type of graph would you use to visualize the distribution of a single variable in Tableau?

A histogram is the most appropriate for showing the distribution of a single continuous variable. In Tableau, drag the variable to Columns and select Histogram from the Show Me panel. For categorical variables, a bar chart showing counts for each category is more appropriate.

62. How can a histogram help in understanding the distribution of a feature like age or petal length?

A histogram shows the frequency of values within bins, revealing the shape (normal, skewed, bimodal), central tendency, and spread of the distribution. For petal length in Iris, it clearly shows distinct clusters corresponding to species. For age in the Adult dataset, it reveals the working-age majority.

63. What insights can be gained by visualizing sepal length vs. petal length in the Iris dataset?

A scatter plot of sepal length vs. petal length with species as color clearly shows that the three Iris species (setosa, versicolor, virginica) form distinct, separable clusters. This indicates that these features are strong discriminators for species classification and that petal dimensions vary significantly across species.

64. Why is 3D visualization useful, and what kind of data does it work best with?

3D visualization is useful for simultaneously analyzing three continuous variables and their interactions. It works best with multivariate data where patterns are not visible in 2D projections, such as scientific measurements, spatial data, or datasets with three interdependent numerical features like age, hours worked, and income.

65. How do you visualize the relationship between multiple variables at once in Tableau?

Use a Scatter Plot Matrix (pairplot) for pairwise relationships between all numeric variables. Add color for categorical grouping (e.g., species or income). Use Tableau's Pages shelf for animated views or FacetGrids (Trellis charts) to split views by category for multidimensional analysis.

66. What is the benefit of using heatmaps or bubble charts for multidimensional data?

Heatmaps encode a third dimension (magnitude) as color intensity on a 2D grid, making it easy to spot high/low value regions across two categorical or discretized variables. Bubble charts encode a third variable as bubble size, effectively showing three dimensions simultaneously without requiring 3D rendering.

67. How would you use a treemap to visualize the distribution of occupations in the Adult dataset?

In Tableau, drag occupation to the Label/Color shelf and count of records to Size. Treemaps use nested rectangles where each rectangle's area is proportional to the count of that occupation. This quickly reveals which occupations are most/least common and how they relate to income groups hierarchically.

68. **Can you explain how a network graph works and what insights it provides in the context of social or scientific data?**

A network graph represents entities as nodes and their relationships as edges (connections). In social data, it reveals communities, influencers, and information flow patterns. In scientific data, it shows co-authorship or gene interaction networks. Node size/color can encode additional attributes like centrality or category.

69. **How could network visualization be applied to relationships in the Iris dataset or demographic groups in the Adult dataset?**

For the Adult dataset, a network graph can visualize relationships between education level, occupation, and income group, showing which education-occupation combinations most strongly correlate with higher income. For Iris, connections between species based on measurement similarity can reveal how closely related species clusters are.

GROUP C: Assignment 8 – Web Scraping (Viva Questions)

70. **What is web scraping, and why is it important for data collection from e-commerce websites?**

Web scraping is the automated extraction of data from websites using programs that simulate browser requests and parse HTML content. It is important for e-commerce data collection because it enables large-scale, real-time gathering of product reviews, prices, and ratings without manual effort, supporting competitive analysis and sentiment analysis.

71. **What are the common challenges when scraping real-time data from websites?**

Common challenges include websites blocking scrapers via CAPTCHAs, IP bans, and rate limiting. Dynamic content loaded via JavaScript cannot be accessed by simple HTTP requests (requiring Selenium). Websites frequently change their HTML structure, breaking existing scrapers, and legal/ethical Terms of Service restrictions must also be considered.

72. **Which Python libraries are commonly used for web scraping, and why would you choose one over another?**

BeautifulSoup is ideal for parsing static HTML pages and is lightweight and easy to use. Scrapy is a full web crawling framework suited for large-scale scraping with built-in pipelines. Selenium is used for JavaScript-heavy dynamic websites since it controls a real browser to render and interact with the page.

73. **Can you explain the role of HTTP requests in web scraping and how they are made in Python?**

HTTP requests fetch the HTML content of a webpage from the server. In Python, the requests library is used: `response = requests.get(url, headers={...})` sends a GET request and the `response.text` or `response.content` contains the HTML that BeautifulSoup parses to extract desired data elements.

74. How do you handle pagination when scraping multiple pages of reviews from an e-commerce site?

Identify the URL pattern for page numbers (e.g., ?page=2) and use a loop to iterate through pages. Check for a 'Next' button or page limit using BeautifulSoup and break the loop when the last page is reached. Add time.sleep() delays between requests to avoid server overload and IP blocking.

75. How would you extract specific elements like customer name, rating, and comments from an e-commerce product page?

Use browser DevTools (Inspect Element) to identify the HTML tags and CSS class names of the target elements. Then use BeautifulSoup methods like soup.find_all('div', class_='review-class') to locate elements and .text.strip() to extract clean text. For ratings, look for numeric text or aria-label attributes in star elements.

76. How do you identify and extract the review tags or keywords from a product review?

Review tags are typically found in or elements with specific CSS classes within the review section. Use soup.find_all() to locate these elements and extract their text. Alternatively, NLP techniques like keyword extraction using NLTK or spaCy can be applied on the review text to identify prominent tags.

77. How do you store the scraped data (e.g., database, CSV, JSON) for further processing?

Use pandas DataFrame to collect scraped records in a list of dictionaries, then export using df.to_csv('reviews.csv', index=False) for CSV or df.to_json('reviews.json') for JSON format. For persistent storage, use SQLite with Python's sqlite3 module or MongoDB for unstructured JSON-like review data.

78. Can you walk me through the steps to set up and execute a basic review scraper for an e-commerce site?

First, inspect the target page's HTML structure to identify review element selectors. Install libraries (pip install requests beautifulsoup4). Write a script: send a GET request, parse HTML with BeautifulSoup, extract review fields using find_all(), store in a list, convert to DataFrame, and export to CSV. Handle pagination and add request delays.